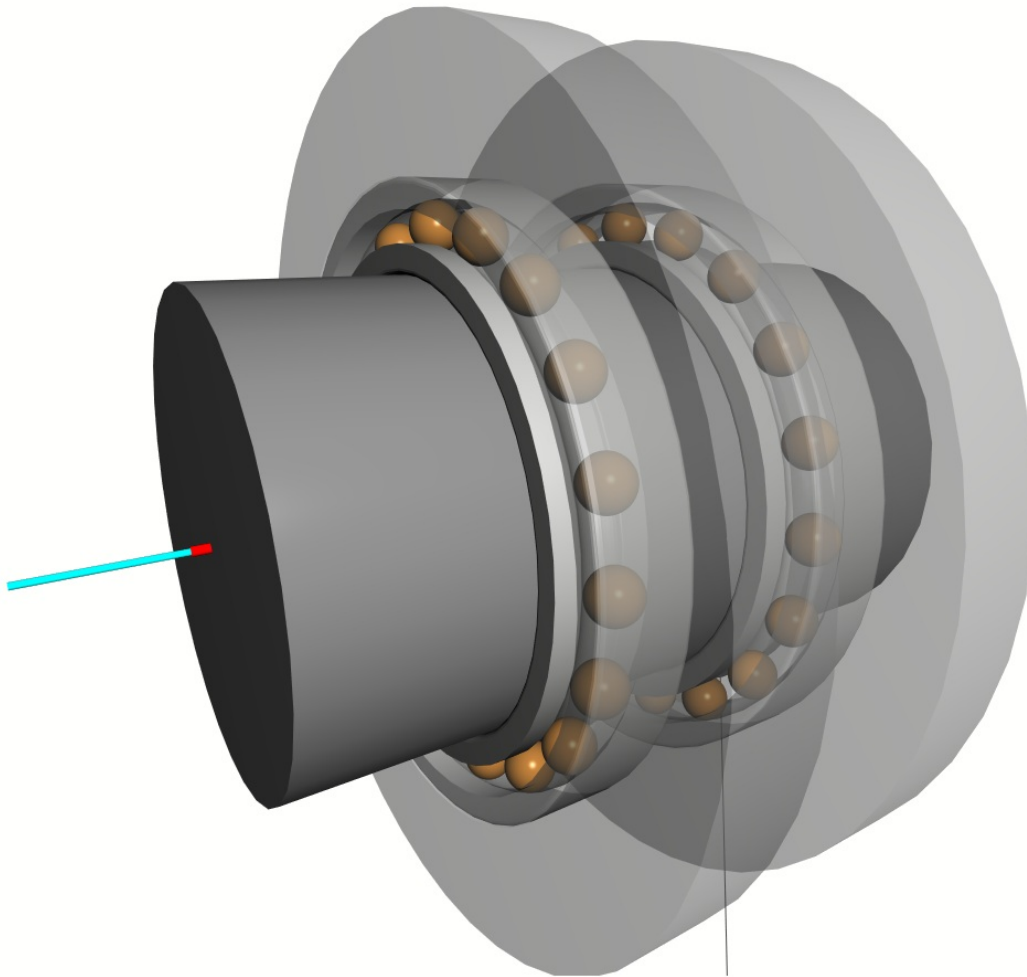


SKF SimPro Quick

61916-2RZ Stiffness 2023 [R DG]



Author: Tyler Yan
Department: Unsprung
Division: LHRE Dynamics
Project number:
Description:
Date: Thu Dec 1 17:58:25 2022
Version: SKF SimPro Quick 4.5

1. Abstract

General comments about bearing rating life:

For rating life results above 100.000 hours, other failure modes than those included in the current rating life models will dominate and limit the life of the bearing.

Life value '0' or 'N/A' is shown if the requirements on either minimum load, permissible misalignment or static safety factor is not met.

If one case in duty cycle does not meet requirements, system life reverts to 0 (or N/A).

For detailed information and assumptions concerning the fatigue life calculation methodology refer to the ISO standard.

Individual bearing life can only be predicted statistically. Life calculations refer only to a bearing population and a given degree of reliability, i.e. 90%. Furthermore field failures are not generally caused by fatigue, but are more often caused by contamination, wear, misalignment and corrosion, or as a result of cage, lubrication or seal failure.

Bearing rating life (ISO)

Bearing	Basic rating life (ISO 281) (L10h) [h]	Modified rating life (ISO 281) (L10mh) [h]	Modified reference rating life (ISO/TS 16281) (L10mrh) [h]
61916-2RZ L	500	1800	1200
61916-2RZ R	300	1000	800

Bearing rating life (SKF load based)

Bearing	SKF load based method (SKF rating life, GBLM) [h]	SKF load based life method	RSF load based [-]
61916-2RZ L	2700	SKF Rating Life	N/A
61916-2RZ R	1600	SKF Rating Life	N/A

SKF load based method (SKF rating life, GBLM): Depending on the bearing design features, the SKF rating life (L10m) or the GBLM load based method (L10GM) is used.

Relative surface fatigue factor (RSF) only valid for bearings with GBLM life

RSF ~ 1: the surface takes most of the fatigue

RSF ~ 0: the surface takes almost no fatigue

2. Input

2.1. Bearing data

Bearing	Bearing designation	Bearing type	Bearing execution	Bore diameter (d) [mm]	Outer diameter (D) [mm]	Bearing width (B) [mm]
61916-2RZ L	61916-2RZ	DGBB	SKF Standard	80.000	110.000	16.000
61916-2RZ R	61916-2RZ	DGBB	SKF Standard	80.000	110.000	16.000

Bearing	Basic dynamic load rating (C) [kN]	Basic static load rating (C0) [kN]	Fatigue load limit (Pu) [kN]	Reference speed [rpm]	Limiting speed [rpm]
61916-2RZ L	25.1	20.4	1.02	12000	6000
61916-2RZ R	25.1	20.4	1.02	12000	6000

2.2. Lubricant data

Lubricant	Viscosity at 40 C [mm2/s]	Viscosity at 100 C [mm2/s]	With effective EP additives	With proven effective EP for etaC<0.2
LGMT 2	110.00	11.00	Off	N/A
MT33	100.00	10.00	Off	N/A

Lubricant	Contamination type ISO Diagram		Filter & oil cleanliness	Oil cleanliness	Grease cleanliness
LGMT 2->ContaminationModel	ISO 281 2007	Grease	N/A	N/A	Grease - normal cleanliness
MT33->ContaminationModel	ISO 281 2007	Grease	N/A	N/A	Grease - normal cleanliness

2.3. Temperatures

Bearing	Shaft / Inner ring [C]	Outer ring / Housing [C]
61916-2RZ L	70	70
61916-2RZ R	70	70

The housing has same temperature as the OR and the shaft same as the IR. The maximum of the IR race and OR race temperatures is used as lubrication temperature.

2.4. Loads & speed

Force	Force [N]				Relative position [mm]			
	X	Y	Z	magnitude	X	Y	Z	magnitude
Force_1	1177	1349	1726	2487	25.40	-203.20	0.00	204.78

Force boundary: magnitude and position relative to the interface in defined coordinate system

Boundary	Rotation speed [rpm]
Rotation speed_1	1366

3. Results

3.1. Loads, static safety & C/P

Bearing	Bearing radial load [N]	Bearing axial load [N]	Forces [N]			Moments [Nm]		
			X	Y	Z	YZ	ZX	XY
61916-2RZ L	7258	888	3001	-6609	888	-1	0	0
61916-2RZ R	8160	840	-1823	7954	840	1	0	0

Bearing	Equivalent static bearing load (P ₀) [N]	Static safety factor s ₀	Equivalent dynamic bearing load (P) [N]	C/P
61916-2RZ L	7258	2.8	7258	3.5
61916-2RZ R	8160	2.5	8160	3.1

Component on the shaft	Position [mm]			Force [N]			Moment [Nm]		
	X	Y	Z	X	Y	Z	YZ	ZX	XY
Shaft system_1->Shaft->intf_61916-2RZ L	0.0	0.0	42.6	-3001	6609	-888	1	0	0
Shaft system_1->Shaft->intf_61916-2RZ R_1	0.0	0.0	82.6	1823	-7954	-840	-1	0	0
Shaft system_1->Shaft->intf_Force_1_1	0.0	0.0	18.3	1178	1345	1728	-351	-44	274
Shaft system_1->Shaft->intf_Rotation speed_1_1	0.0	0.0	62.3	0	0	0	0	0	0
Shaft system_1->Shaft->intf_Torque reaction_1_1	0.0	0.0	123.4	0	0	0	0	0	-274

Bearing stiffness

Bearing	Elements	Bearing stiffness				
		1: X [1/m]	2: Y [1/m]	3: Z [1/m]	4: YZ [1/rad]	5: XZ [1/rad]
61916-2RZ L	1: F _x [N]	3.70E08	-4.73E07	-2.18E07	-8.76E04	-3.48E06
	2: F _y [N]	-4.73E07	4.52E08	4.80E07	3.63E06	8.76E04
	3: F _z [N]	-2.18E07	4.80E07	7.96E07	9.57E05	4.35E05
	4: M _{yz} [Nm]	-8.77E04	3.63E06	9.57E05	1.07E05	1.92E04
	5: M _{xz} [Nm]	-3.48E06	8.77E04	4.35E05	1.92E04	7.31E04
61916-2RZ R	1: F _x [N]	3.56E08	-2.86E07	1.17E07	-3.72E04	-3.34E06
	2: F _y [N]	-2.86E07	4.74E08	-5.10E07	3.49E06	3.72E04
	3: F _z [N]	1.17E07	-5.10E07	8.04E07	-9.27E05	-2.12E05
	4: M _{yz} [Nm]	-3.72E04	3.49E06	-9.27E05	1.15E05	1.18E04
	5: M _{xz} [Nm]	-3.34E06	3.72E04	-2.12E05	1.18E04	6.65E04

Bearing stiffness is defined as the contact stiffness of the IR in relation to the OR, in the actual IR coordinate system. Please note that the stiffness is derived for constant displacement and not for constant load. Be aware that ring stiffness is not included in the bearing stiffness matrix (only contact stiffness is considered).

Bearing contact data

Bearing	RE RE position [deg]	Load IR [N]	Angle IR [deg]	Def. IR [μm]	Load OR [N]	Angle OR [deg]	Def. OR [μm]
Shaft system_1->61916-2RZ L->ReDGBB_1	1	0	464	8.8	6	465	8.7
	2	19	157	13.5	3	157	13.5
	3	38	41	18.7	1	42	18.5
	4	57	30	23.5	1	31	23.1
	5	76	74	27.0	2	75	26.9
	6	95	144	29.4	3	144	29.4
	7	114	180	30.3	3	180	30.3
	8	133	148	29.5	3	149	29.5

	9	152	79	27.3	2	80	27.2	2
	10	171	32	23.8	1	33	23.5	1
	11	189	38	19.1	1	38	18.8	1
	12	208	142	13.9	3	142	13.9	3
	13	227	433	9.1	5	433	9.1	7
	14	246	917	5.1	9	917	5.0	12
	15	265	1450	2.3	12	1451	2.2	16
	16	284	1809	0.8	14	1810	0.6	19
	17	303	1824	0.7	14	1825	0.6	19
	18	322	1488	2.1	12	1488	2.0	17
	19	341	959	4.9	9	960	4.8	12
Shaft system_1->61916-2RZ R->ReDGBB_1	1	0	37	18.7	1	37	18.4	1
	2	19	169	13.3	3	170	13.3	4
	3	38	520	8.5	6	521	8.4	8
	4	57	1078	4.5	10	1078	4.4	13
	5	76	1665	1.7	13	1666	1.5	18
	6	95	2030	0.3	15	2031	0.2	20
	7	114	2001	0.4	15	2002	0.3	20
	8	133	1592	2.0	13	1593	1.9	17
	9	152	994	4.9	10	995	4.8	13
	10	171	458	9.1	6	458	9.0	8
	11	189	140	14.0	3	141	14.0	3
	12	208	30	19.4	1	31	19.1	1
	13	227	24	24.3	1	24	23.9	1
	14	246	69	27.8	2	70	27.7	2
	15	265	138	30.1	3	139	30.1	3
	16	284	169	30.8	3	169	30.9	4
	17	303	130	29.9	2	130	29.9	3
	18	322	61	27.4	1	61	27.3	2
	19	341	21	23.8	1	22	23.3	1

- Position: angular position of rolling elements (from X-axis towards Y-axis)
- Load: rolling element contact loads
- Angle: the absolute contact angle
- Def: deformation of contacts

Bearing ring displacement local

Bearing	Displacement inner ring [um]			Displacement outer ring [um]		
	X	Y	Z	X	Y	Z
Shaft system_1->61916-2RZ L	18	-39	115	0	0	0
Shaft system_1->61916-2RZ R	-10	45	115	0	0	0

Displacements in the local coordinate system

- The displacement of the ring is the weighted average of the raceway displacements. The centre of the bearing unit is used as reference. The raceways closer to the centre of the bearing unit have a higher weight. This also applies to bearings with split inner ring or outer ring.

- Use output of 'Interface displacements' for displacements of individual interfaces (like bore or outer diameter).

3.2. Bearing contact data

Bearing	Max pressure (IR) [N/mm2]	Max pressure (OR) [N/mm2]
61916-2RZ L	2442	2936
61916-2RZ R	2531	3043

3.3. Bearing clearance

Bearing	Internal radial clearance before mounting [um]	Operating radial clearance [um]	Internal axial clearance before mounting [um]	Operating axial clearance [um]
61916-2RZ L	20	20	186	186
61916-2RZ R	20	20	186	186

Estimated operational clearance/preload values are shown. These can differ slightly from the actual operational radial clearance or preload in the final analysis.
 Considers only a single bearing.
 The elongation or displacements of the shaft and housing are not taken into account.

3.4. Relubrication interval & grease life

Bearing	Lubricant	Catalogue relubrication interval [h]	Catalogue grease life [h]	Grease relubrication quantity from the side [gr]	Grease relubrication quantity through centre of the bearing [gr]
61916-2RZ L	MT33	N/A	N/A	N/A	N/A
61916-2RZ R	MT33	N/A	N/A	N/A	N/A

Grease life is calculated based on the calculation of the relubrication interval.
 The limitation of the grease life to 100,000 hours only happens after possible correction factors have been applied.

3.5. Lubrication conditions

Bearing	Lubricant	Bearing speed [rpm]	ndm factor [mm/min]	Lub. temperature [C]	Actual viscosity [mm ² /s]	Kappa	etaC
61916-2RZ L	MT33	1366	129770	70	25.4	2.47	0.73
61916-2RZ R	MT33	1366	129770	70	25.4	2.47	0.73

3.6. Bearing rating life

General comments about bearing rating life:

For rating life results above 100.000 hours, other failure modes than those included in the current rating life models will dominate and limit the life of the bearing.

Life value '0' or 'N/A' is shown if the requirements on either minimum load, permissible misalignment or static safety factor is not met.

If one case in duty cycle does not meet requirements, system life reverts to 0 (or N/A).

For detailed information and assumptions concerning the fatigue life calculation methodology refer to the ISO standard.

Individual bearing life can only be predicted statistically. Life calculations refer only to a bearing population and a given degree of reliability, i.e. 90%. Furthermore field failures are not generally caused by fatigue, but are more often caused by contamination, wear, misalignment and corrosion, or as a result of cage, lubrication or seal failure.

Bearing rating life (ISO)

Bearing	Life factor aISO	Basic rating life (ISO 281) (L10h) [h]	Modified rating life (ISO 281) (L10mh) [h]	Modified reference rating life (ISO/TS 16281) (L10mrh) [h]
61916-2RZ L	3.65	500	1800	1200
61916-2RZ R	3.09	300	1000	800

Bearing rating life (SKF load based)

Bearing	Life factor aSKF	SKF load based method (SKF rating life, GBLM) [h]	SKF load based life method RSF load based [-]
61916-2RZ L	5.47	2700	SKF Rating Life N/A
61916-2RZ R	4.53	1600	SKF Rating Life N/A

SKF load based method (SKF rating life, GBLM): Depending on the bearing design features, the SKF rating life (L10m) or the GBLM load based method (L10GM) is used.

Relative surface fatigue factor (RSF) only valid for bearings with GBLM life

RSF ~ 1: the surface takes most of the fatigue

RSF ~ 0: the surface takes almost no fatigue

3.7. Bearing frictional moment & power loss

Bearing	Total frictional moment [Nmm]	Starting torque [Nmm]	Friction torque sources [Nmm]				Power loss [W]
			1) Rolling resistance	2) Sliding	3) Seal	4) Drag	
61916-2RZ L	602	1002	266	336	0	0	86
61916-2RZ R	680	1218	272	408	0	0	97

Based on the mean bearing raceway temperature (average of inner ring and outer ring).
Starting torque: At start 20-30 degC and zero speed

3.8. Bearing frequencies

Bearing	Rotational frequency IR [hertz]	Rotational frequency OR [hertz]	Rotational frequency of RE set and cage [hertz]	Rolling element about its axis [hertz]	Over-rolling frequency of point on IR [hertz]	Over-rolling frequency of point on OR [hertz]	Over-rolling frequency of point on RE [hertz]
61916-2RZ L	22.77	0	10.34	122.82	236.09	196.48	245.64
61916-2RZ R	22.77	0	10.34	122.82	236.11	196.46	245.64

3.9. Bearing & shaft displacement

Bearing	Displacement [um]			Misalignment [min]			Total misalignment [min]
	X	Y	Z	YZ	ZX	XY	
61916-2RZ L	18	-39	115	7	3	0	8
61916-2RZ R	-10	45	115	-8	-2	0	8

- Bearing displacement and misalignment of inner ring relative to outer ring
- The displacements and misalignments are displayed in the local coordinate system of the bearing

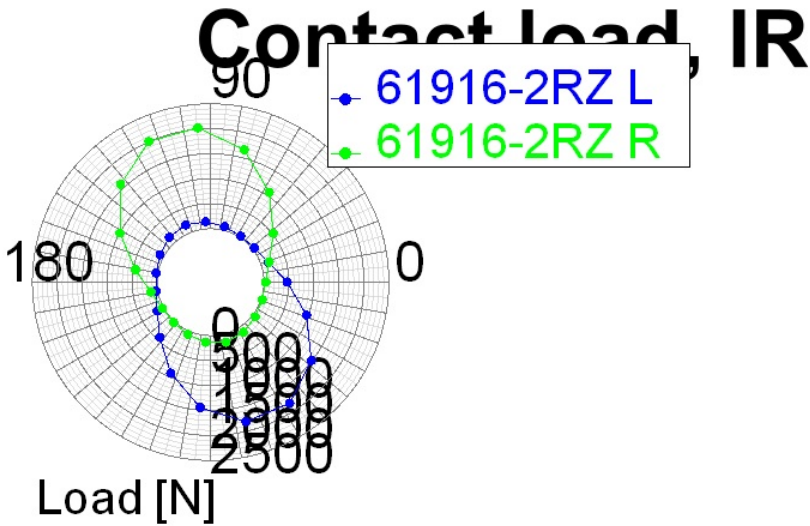
Shaft displacement & misalignment (at interfaces)

Shaft position	Displacement [um]			Misalignment [min]		
	X	Y	Z	YZ	ZX	XY
intf_61916-2RZ L	18	-39	115	-7.4	-2.5	0.2
intf_61916-2RZ R_1	-10	45	115	-7.3	-2.4	0.1
intf_Force_1_1	35	-92	115	-7.4	-2.5	0.3
intf_Rotation speed_1_1	4	2	115	-7.3	-2.4	0.2
intf_Torque reaction_1_1	-39	132	114	-7.3	-2.4	0.0

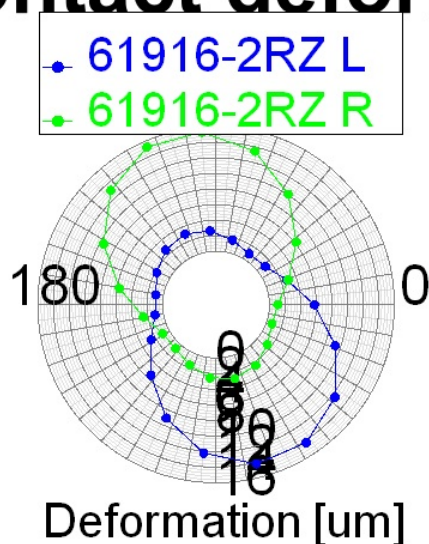
Shaft displacement & misalignment (at positions)

Shaft	Node	Axial position [mm]	Displacement [um]			Misalignment [min]		
			X	Y	Z	YZ	ZX	XY
Shaft	1	0.00	49	-131	115	-7.4	-2.5	0.3
	2	18.30	35	-92	115	-7.4	-2.5	0.3
	3	42.61	18	-39	115	-7.4	-2.5	0.2
	4	62.25	4	2	115	-7.3	-2.4	0.2
	5	82.61	-10	45	115	-7.3	-2.4	0.1
	6	102.99	-25	89	114	-7.3	-2.4	0.1
	7	123.38	-39	132	114	-7.3	-2.4	0.0
	8	124.50	-40	134	114	-7.3	-2.4	0.0

3.10.



Contact deformation, IR



Contact angle, IR

